

Data and group coursework

SMM048 (2025-26) Group Coursework 01 - Group 07

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11 3 2026

Contents

1	Factor Analysis	2
1.1	Figures and tables for Q1	3
2	Copula analysis	6
2.1	Figures and tables for Q2	7
3	Principal Component Analysis	9
3.1	Figures and tables for Q3	10
4	Time series	15
4.1	Figures and tables for Q4	16
5	Appendix	22
5.1	bonus : Q1 Factor analysis comparison(number of factors 5 vs 6)	22
5.2	bonus : Q2 Copula	23
5.2.1	Modelling using Poisson and Tweedie margin	23
5.2.2	Goodness-of-fit tests	23
5.2.3	Adding covariates into the models	23
5.3	bonus : Q3 Principal Component Analysis	24

1 Factor Analysis

An exploratory factor analysis (EFA) was conducted on the 25 personality items measured on a 6-point Likert scale. Missing values (NA) were handled using listwise deletion. In addition, several negatively worded items (X1: empathy, X9: carelessness, X10: idleness, X11: quietness, X12: aloofness, X22: avoidance, and X25: superficiality) were reverse-coded so that higher values consistently represented higher levels of the underlying traits.

To determine the appropriate number of factors, parallel analysis, principal component analysis (PCA) in Figure 1, and the scree plot in Figure 2 were examined. The scree plot shows a sharp decline in eigenvalues for the first components (5.13, 2.75, 2.14, 1.85, 1.55, and 1.07). According to the Kaiser criterion (eigenvalue > 1), up to six components could be retained. However, the sixth eigenvalue is only slightly above the threshold, and the scree plot levels off after the fifth component, suggesting that additional components contribute little to explaining the overall variance. Considering these results, a five-factor solution was selected. For completeness, the comparison between the five- and six-component solutions obtained using PCA is provided in the Appendix.

After determining the number of factors, the factor loading matrices were estimated and rotated to improve interpretability. Figure 3 presents the rotated factor loading matrices obtained using both varimax and promax rotations. The two rotation methods produce broadly similar loading patterns, suggesting that the underlying factor structure is relatively stable. In most cases, items load strongly on a single factor, while cross-loadings across multiple factors remain limited. The promax rotation allows correlations between factors and produces slightly clearer clustering of items, although the overall interpretation of the factor structure remains consistent across the two rotation methods.

The uniqueness values reported in Table 1 ranges from 0.27 to 0.83, with an average of approximately 0.58. This indicates that the extracted factors explain a moderate proportion of the variance in the observed items. The rotated factor loadings reveal a clear five-factor structure consistent with widely recognised personality dimensions. **Factor 1** loads strongly on items related to **emotional instability** (X16–X20), suggesting that this factor represents neuroticism. **Factor 2** reflects **extraversion**, with items describing sociability and interpersonal engagement (X11–X15). **Factor 3** corresponds to **conscientiousness**, capturing traits such as organisation, diligence, and careful work behaviour (X6–X10). **Factor 4** represents **agreeableness**, including items related to empathy, cooperation, and concern for others (X2–X5). Finally, **Factor 5** appears to capture **openness**-related characteristics, including curiosity, imagination, and intellectual engagement (X21–X23, X25). Overall, the factor analysis results provide empirical support for a five-factor personality structure within the dataset.

$$\begin{aligned}
 \hat{f}_1: \text{emotional instability} &\approx 0.82x_{16} + 0.79x_{17} + 0.71x_{18} + 0.56x_{19} + 0.52x_{20} \\
 \hat{f}_2: \text{extraversion} &\approx 0.59x_{11} + 0.67x_{12} + 0.49x_{13} + 0.61x_{14} + 0.49x_{15} \\
 \hat{f}_3: \text{conscientiousness} &\approx 0.53x_6 + 0.62x_7 + 0.55x_8 + 0.65x_9 + 0.57x_{10} \\
 \hat{f}_4: \text{agreeableness} &\approx 0.60x_2 + 0.66x_3 + 0.45x_4 + 0.58x_5 \\
 \hat{f}_5: \text{openness} &\approx 0.52x_{21} + 0.45x_{22} + 0.61x_{23} + 0.51x_{25}
 \end{aligned}$$

Table 3 presents the rotated factor loadings obtained from the exploratory factor analysis, showing the relationship between the observed variables and the five latent factors. Larger absolute loading values indicate a stronger relationship between an observed variable and the corresponding latent factor. Using item X16 as an example, the loading for Factor 1 is 0.82, while the loadings on the other factors are close to zero, indicating that this item is primarily associated with Factor 1. This suggests that X16 strongly reflects the latent dimension represented by Factor 1. Since Factor 1 is characterised by items associated with emotional instability (X16–X20), it can be interpreted as representing the neuroticism dimension.

$$X_{16} = 0.82\text{Factor1} + 0.09\text{Factor2} - 0.04\text{Factor3} - 0.21\text{Factor4} - 0.08\text{Factor5} + e_{16}$$

1.1 Figures and tables for Q1

Parallel analysis suggests that the number of factors = 6 and the number of components = 5

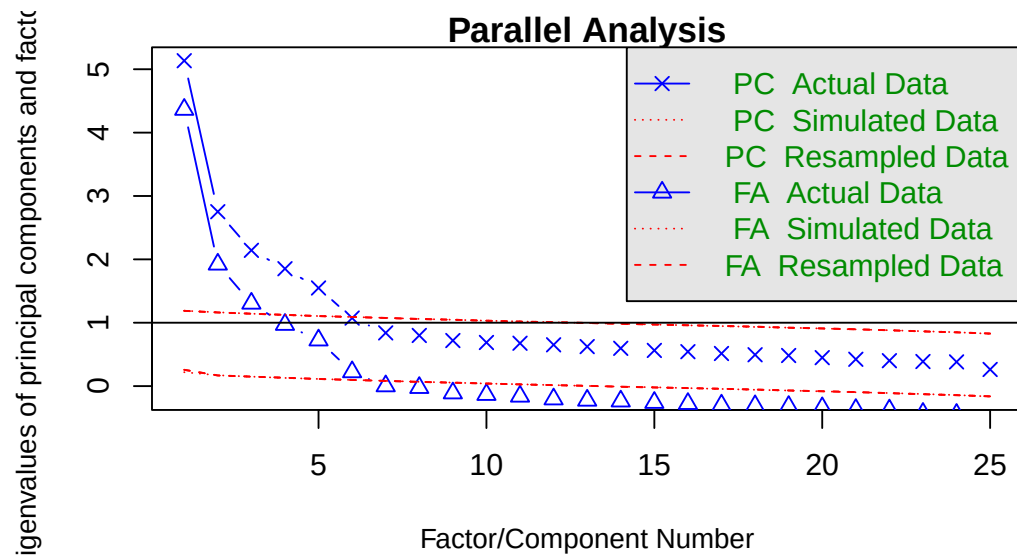


Figure 1: parallel analysis for determining the number of factors

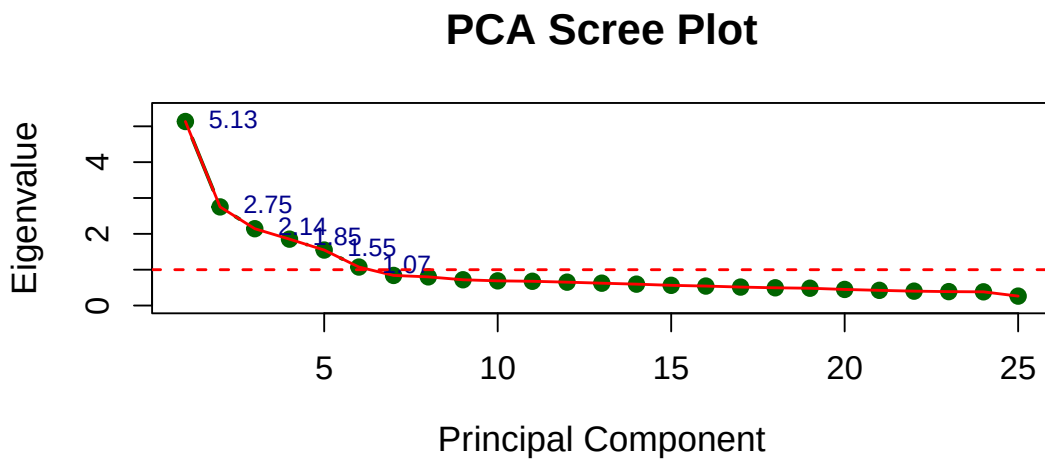


Figure 2: Scree plot for determining the number of factors

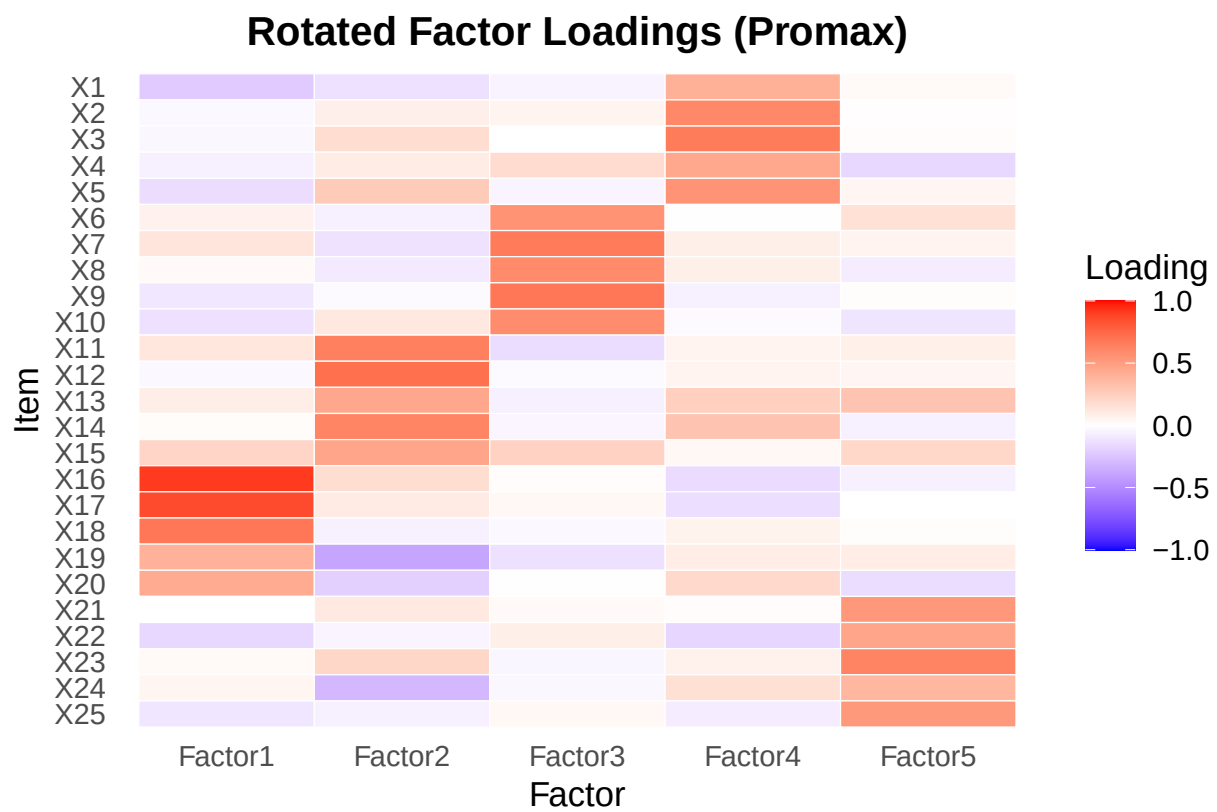
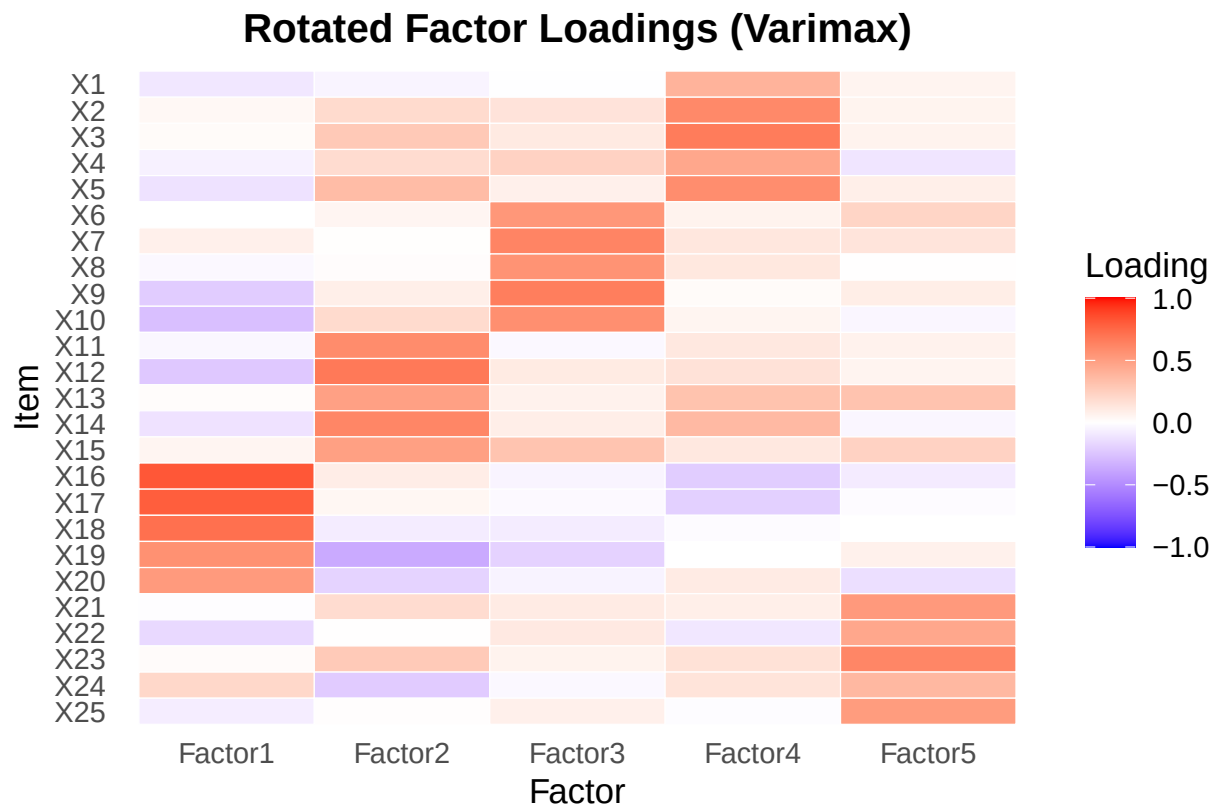


Figure 3: Rotated Factor loadings(Varimax vs Promax)

Table 1: Summary of Uniqueness Values

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.271	0.507	0.569	0.577	0.675	0.83

Table 2: Rotated Factor Loadings for 5 Factors ($|\text{loading}| \geq 0.40$)

Factor	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	X12	X13	X14	X15	X16	X17	X18	X19	X20	X21	X22	X23	X24	X25
F1																0.82	0.79	0.71	0.56	0.52					
F2											0.59	0.67	0.49	0.61	0.49										
F3						0.53	0.62	0.55	0.65	0.57															
F4		0.60	0.66	0.45	0.58																				
F5																					0.52	0.45	0.61		0.51

Table 3: Rotated Factor Loadings

	Factor1	Factor2	Factor3	Factor4	Factor5
X1	-0.10	-0.05	0.00	0.39	0.06
X2	0.04	0.19	0.14	0.60	0.06
X3	0.02	0.28	0.11	0.66	0.06
X4	-0.06	0.18	0.23	0.45	-0.11
X5	-0.12	0.35	0.08	0.58	0.08
X6	0.00	0.05	0.53	0.06	0.22
X7	0.08	0.01	0.62	0.13	0.14
X8	-0.03	0.01	0.55	0.12	0.00
X9	-0.22	0.08	0.65	0.02	0.09
X10	-0.27	0.19	0.57	0.05	-0.04
X11	-0.03	0.59	-0.03	0.12	0.07
X12	-0.23	0.67	0.11	0.15	0.06
X13	0.02	0.49	0.07	0.31	0.31
X14	-0.12	0.61	0.09	0.36	-0.04
X15	0.05	0.49	0.31	0.12	0.23
X16	0.82	0.09	-0.04	-0.21	-0.08
X17	0.79	0.04	-0.02	-0.20	-0.02
X18	0.71	-0.08	-0.08	-0.02	0.00
X19	0.56	-0.37	-0.19	0.00	0.07
X20	0.52	-0.19	-0.05	0.11	-0.14
X21	-0.01	0.18	0.10	0.09	0.52
X22	-0.16	0.00	0.11	-0.10	0.45
X23	0.02	0.28	0.07	0.15	0.61
X24	0.21	-0.22	-0.03	0.14	0.37
X25	-0.08	0.01	0.08	-0.01	0.51

2 Copula analysis

As the majority of claim amounts are zero in the dataset (approximately 93.2% of 67,856 records), different approaches were initially considered but not pursued. Please refer to `Modelling using Poisson and Tweedie` margin section in the appendix.

Several truncated count models were considered to account for the absence of zero claims in the dataset, as shown in table 4. The **Truncated Poisson** model is preferred, having the lowest BIC and therefore the most parsimonious. Although the truncated Poisson–Inverse Gaussian specification produces the smallest AIC, the improvement is negligible ($\Delta\text{AIC} \approx 1.13$). The selected model assumes that claims occur independently at a constant rate over the exposure period, with an estimated annual claim rate of $\hat{\mu} = e^{-2.66711} = 0.0694$ indicating that claims arrive relatively infrequently in the portfolio. Claim severity was modelled using positive claim amounts only. Distribution is strongly right-skewed, indicating a small proportion of claims have extremely high severity. Among the models tested, the **Inverse Gaussian** distribution provides the best fit, achieving the lowest AIC and BIC values, as shown in table 5. It has an average positive claim severity of approximately $\hat{\mu} = e^{7.60808} = 2011$, with variability increasing with the cube of the mean, consistent with the heavy right tail commonly observed in severity data. As an extension, we have also attempted to include covariates to model the marginal distribution parameters, but we did not see much improvement in the model AIC and BIC. For more details, please refer to `Adding covariates into the models` section in the appendix.

Both graphical and statistical methods are used for model diagnostic. The residual quantile Q–Q plots (figure 4) show that the truncated Poisson model provides a satisfactory representation of claim frequency, while the inverse Gaussian distribution captures the overall shape of the severity distribution. Although the lower tail appears to deviate from the diagonal line, indicating that the model underestimates very small claim amounts, the overall fit is considered adequate for modelling dependence in the following copula analysis. Convergence diagnostics also indicate stable estimation. In both marginal models the maximum absolute gradient is close to zero and the observed information matrix is positive definite, confirming that the likelihood optimisation has successfully converged. Additional goodness-of-fit tests analysis we have done is discussed in `Goodness-of-fit Tests` section in the appendix.

To model the dependence between claim frequency and claim severity, four copula families were considered: Gaussian, Student-t, Clayton and Gumbel. The results are presented in table 6. Among these candidates, the Clayton copula provides the best fit, achieving the lowest AIC and BIC values by a clear margin. Convergence diagnostics confirm stable estimation, with a maximum gradient of 8.45×10^{-6} and a positive definite information matrix. The estimated Clayton copula parameter is positive and statistically significant $\theta = 1.47$, indicating positive dependence between claim frequency and claim severity. It corresponds to a Kendall's $\tau = \frac{\theta}{(\theta+2)} = 0.4236$, which indicates a moderate positive association between claim frequency and claim severity. We therefore fit **Clayton copula** with `claytonCopula(param = 1.47, dim = 2)`. for pseudo observation plot: The pseudo-observations are obtained by transforming claim frequency and severity to uniform margins (0,1). The vertical bands arise because claim frequency is discrete, with most observations corresponding to one or two claims. Claim severity ranks are mainly spread across the mid-to-upper range of U2, indicating many observations with moderate claim costs rather than strong clustering in the extreme tails. The plot suggests a mild positive association, as higher claim frequencies tend to correspond to higher severity ranks. This pattern supports the use of a positively dependent copula such as the Clayton copula, while providing little visual evidence of strong upper-tail clustering that would favour a Gumbel copula. for scatter plot: The simulated scatter plot shows simulated observations from the fitted Clayton copula with Kendall's $\tau = 0.4236$. It exhibits positive dependence with noticeable concentration in the bottom left region of the unit square, reflecting the lower-tail dependence captured by the Clayton copula. This implies that policies with relatively few claims are more likely to be associated with relatively small claim costs. Such asymmetric dependence is consistent with the empirical pattern observed in Figure X and contrasts with copulas such as the Gumbel copula, which emphasise upper-tail dependence that is not strongly supported by the data, meaning that extremely high claim counts are not systematically associated with extremely large claim amounts.

2.1 Figures and tables for Q2

Table 4: Claim frequency model performance

Models	df	AIC	BIC
Truncated Poisson	1	2297.331	2303.771
Truncated NB I	1	2296.227	2309.105
Truncated Poisson Inverse Gaussian	2	2296.119	2309.077

Table 5: Claim severity model performance

Models	df	AIC	BIC
LogNormal	2	77708.31	77721.19
Gamma	2	79329.84	79342.72
Inverse Gaussian	2	77187.65	77200.53

Table 6: Copula model performance

Models	df	AIC	BIC
Gaussian	4	79384.86	79410.61
Student-t	4	79590.55	79616.30
Clayton	4	79321.95	79616.30
Gumbel	4	79482.90	79508.66

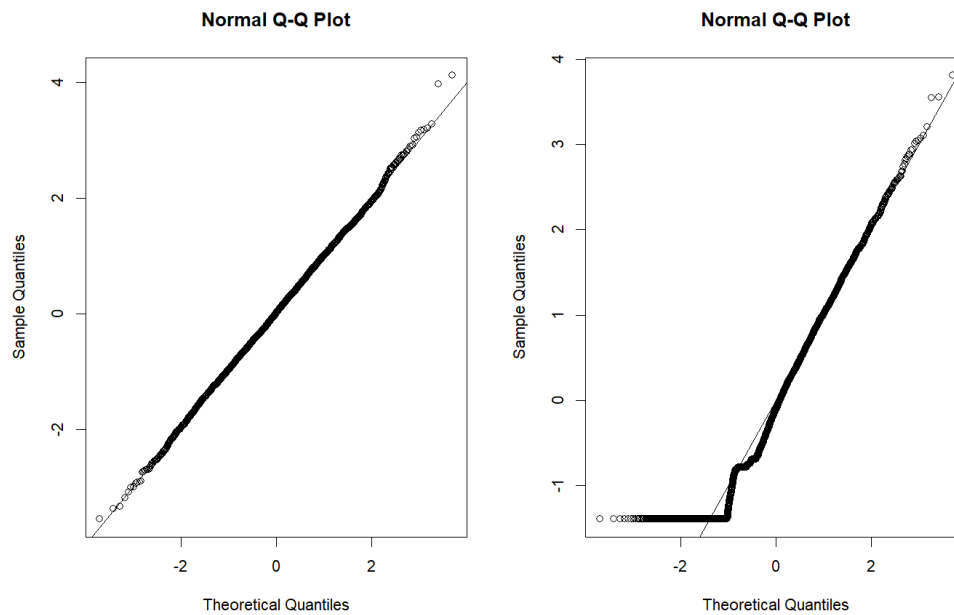


Figure 4: Residual quantile Q-Q plot of the fitted truncated Poisson for the claim marginal distribution (left) and Inverse Gaussian model for severity marginal distribution (right).

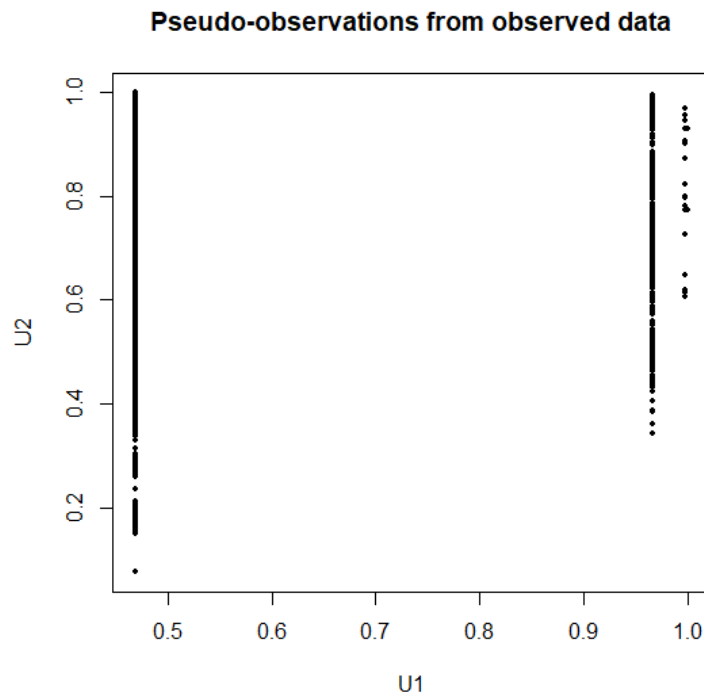


Figure 5: Observed copula

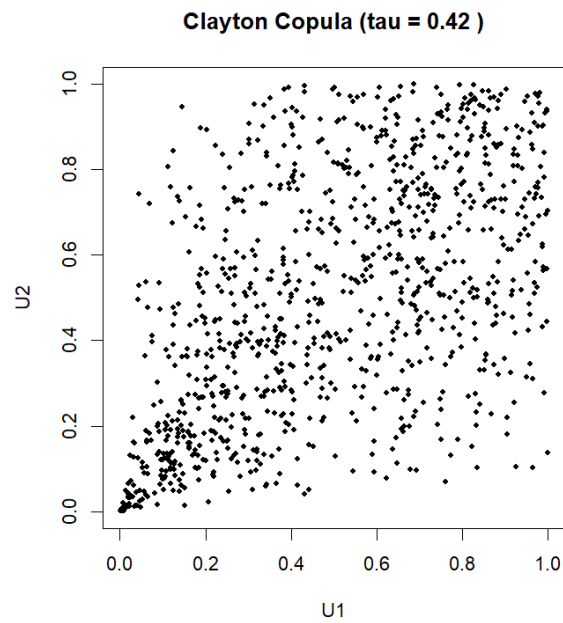


Figure 6: A plot of points generated from Clayton copula. $U1$ and $U2$ are the quantile of the first and second distribution respectively.

3 Principal Component Analysis

We applied Principal Component Analysis (PCA), an unsupervised dimensionality-reduction method, to transform the 11 variables into a smaller set of uncorrelated components that successively explain the maximum possible variance in the dataset. The aim of this approach is to summarise the information contained in a large number of variables while reducing redundancy caused by correlations among them. Variables were standardised using z-scores before PCA because the features are measured in different units, such as income, age, and rating scores. Without scaling, variables with larger variances may dominate PC1 due to measurement scale rather than the true underlying structure in the data, potentially leading to misleading interpretations.

The correlation structure of the variables is illustrated in Figure 7. The correlation matrix shows generally weak to moderate correlations among variables, suggesting that while some relationships exist, no single pair of variables dominates the structure of the dataset. However, several variables related to purchasing behaviour, such as spending score, shopping frequency, and brand loyalty, are positively correlated, indicating that customers who shop frequently also tend to have higher spending levels and stronger brand preferences.

To determine the appropriate number of components, the scree plot shown in Figure 8 was examined. The scree plot indicates a clear elbow after the third component, suggesting that the marginal contribution of additional components becomes relatively small beyond this point. Although the Kaiser criterion suggests retaining four components (eigenvalue > 1), the elbow criterion and the interpretability of the rotated components support retaining three principal components.

The proportion of variance explained by each component is shown in Figure 9, while the cumulative variance explained is presented in Figure 10. The first three principal components explain approximately 55% of the total variance, which is sufficient to summarise the main behavioural patterns in the data. Parallel analysis in Figure 11 was also examined to determine the appropriate number of components. Although the test suggests retaining two components, the third component still contributes meaningful interpretability and increases the cumulative variance explained. Therefore, three components were retained for further interpretation.

The rotated loading structure obtained using varimax rotation is illustrated in Figure 12 and 13 and summarised in Table 7. The rotated loading plot suggests that **RC1** is mainly driven by **purchasing behaviour**, with income, brand loyalty, shopping frequency, and spending score loading strongly on this component, indicating a dimension of purchasing intensity. **RC2** is primarily associated with **age-spend contrast**, indicating a contrast between younger high-spending customers and older customers, noting that the sign of loadings is arbitrary and does not affect interpretation. **RC3** captures variation in **marketing channel engagement**: promotion usage and email engagement load positively, whereas social media engagement loads negatively, indicating different preferences across communication channels. Overall, the three rotated components summarise customer behaviour into purchasing intensity, demographic influence, and marketing responsiveness.

In Figure 14, The scatter plot shows the distribution of observations in the space of the first two principal components, with clusters identified using k-means clustering. The results suggest three groups of customers with different behavioural patterns. One group is associated with higher purchasing intensity (high PC1 values), while another group corresponds to lower purchasing activity. The third group lies between these extremes, representing customers with moderate purchasing behaviour. The number of clusters was set to three to provide a simple segmentation of customers in the reduced component space. This choice is supported by the visual separation observed in the scatter plot and facilitates the interpretation of distinct behavioural patterns among customers.

Overall, these components provide a compact representation of the main behavioural patterns in the dataset and facilitate interpretation of customer segmentation. By reducing the dimensionality of the data while retaining its most important information, PCA helps reveal the underlying structure of customer behaviour and supports more meaningful analysis of purchasing patterns.

3.1 Figures and tables for Q3

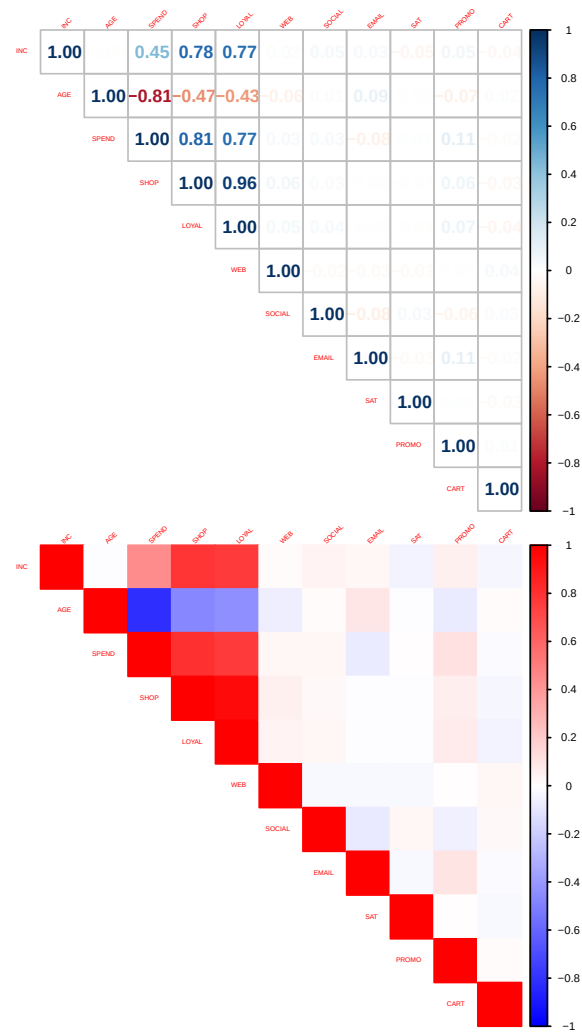


Figure 7: Correlation matrix (numbers, heatmap)

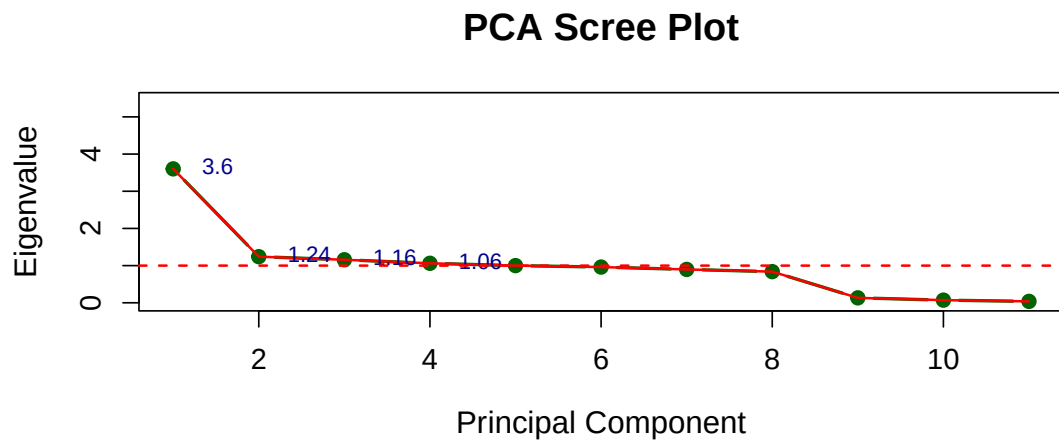


Figure 8: Scree plot

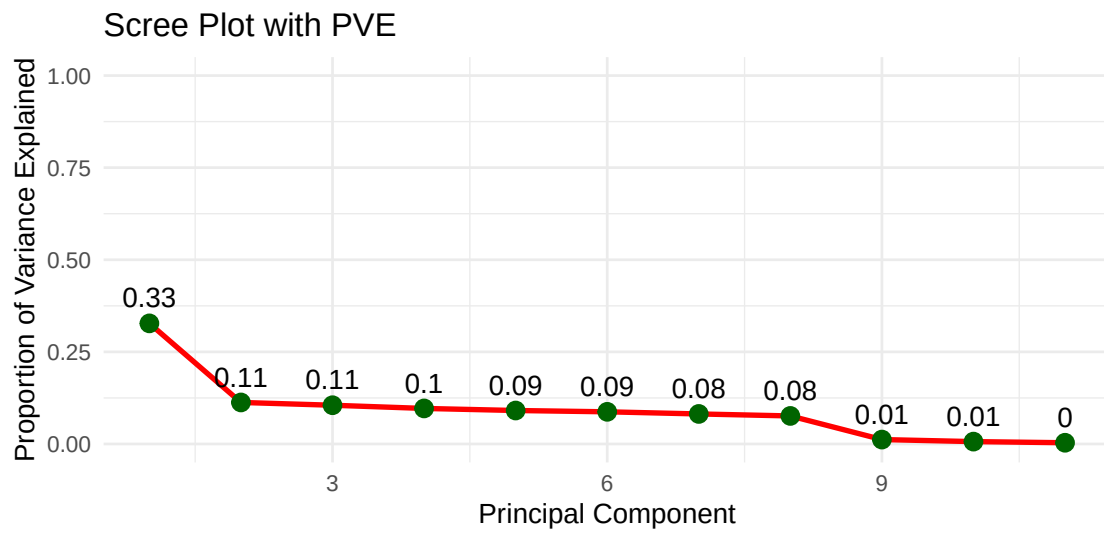


Figure 9: Scree Plot with PVE

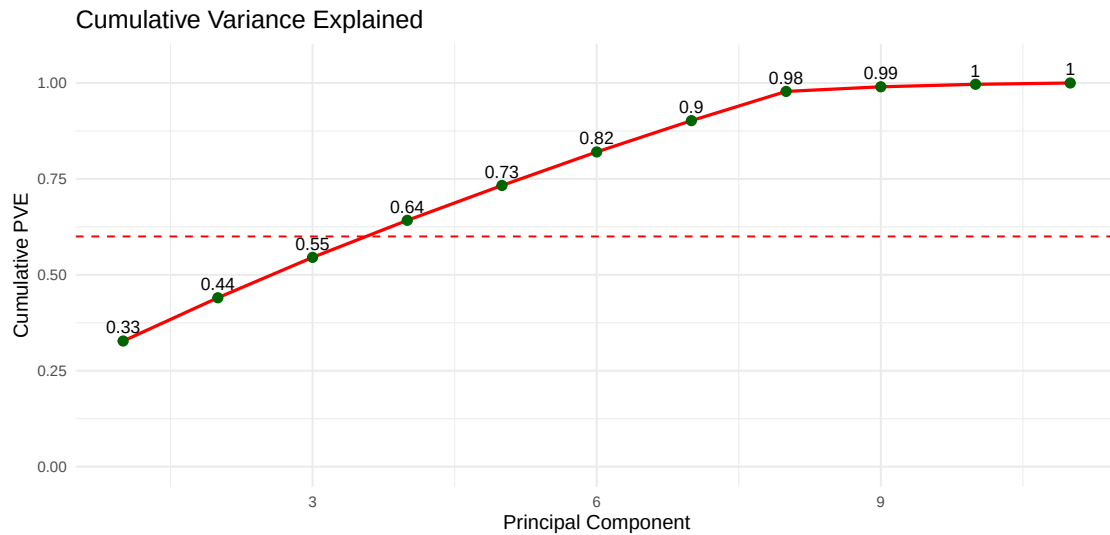


Figure 10: Cumulative variance

Parallel analysis suggests that the number of factors = 3 and the number of components = 3

eigenvalues of principal components and factor analysis

Parallel analysis

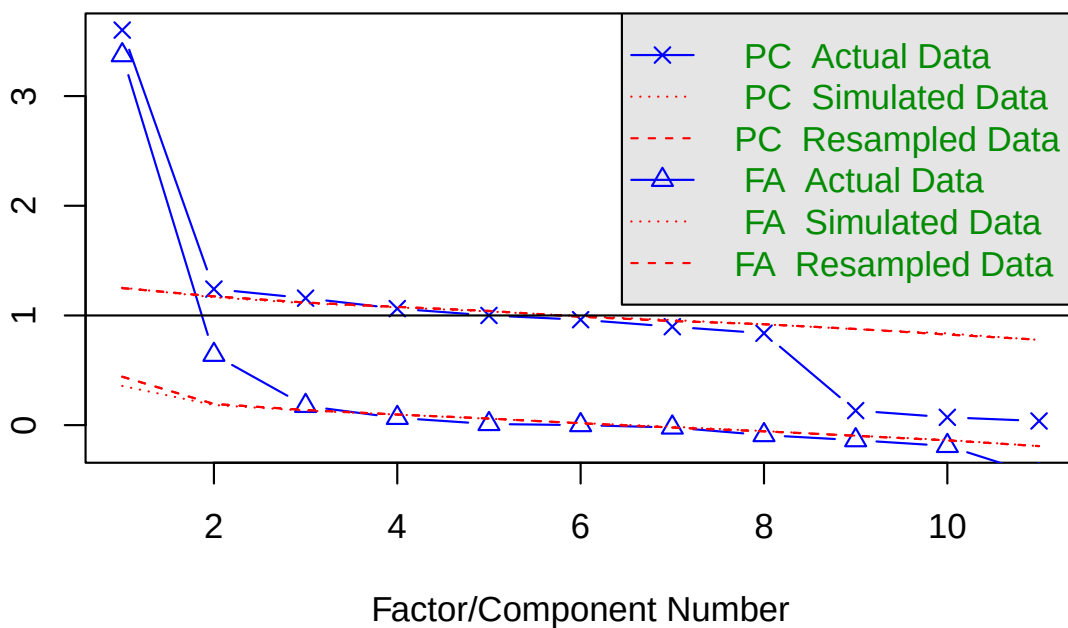


Figure 11: parallel analysis.

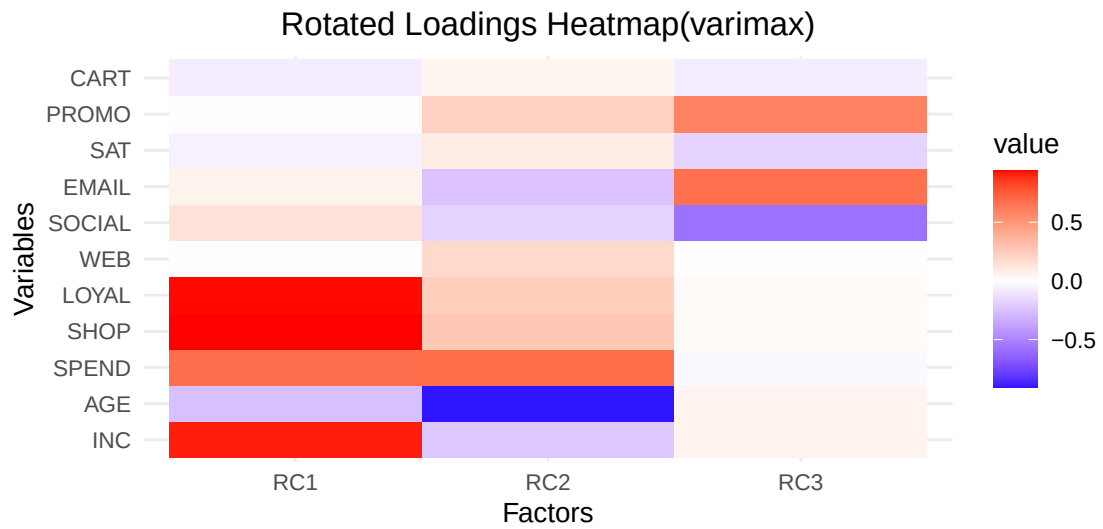


Figure 12: Rotated loading heatmap.

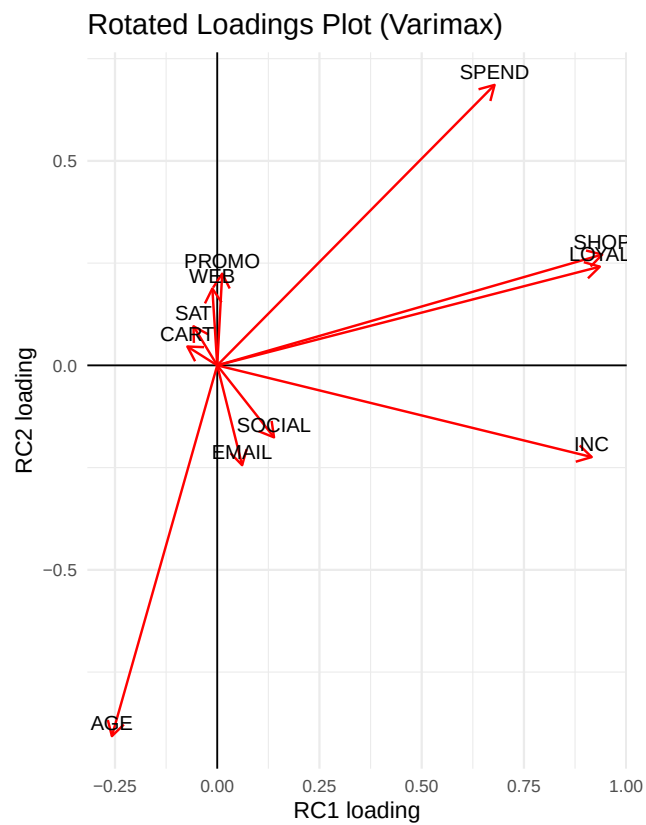


Figure 13: loading plot of the rotated PCA using varimax rotation.

Table 7: Rotated PCA loadings after varimax rotation ($|\text{loading}| \geq 0.40$)

Factor	INC	AGE	SPEND	SHOP	LOYAL	WEB	SOCIAL	EMAIL	SAT	PROMO	CART
RC1	0.92		0.68	0.94	0.94						
RC2		-0.91	0.69								
RC3							-0.58	0.67		0.59	

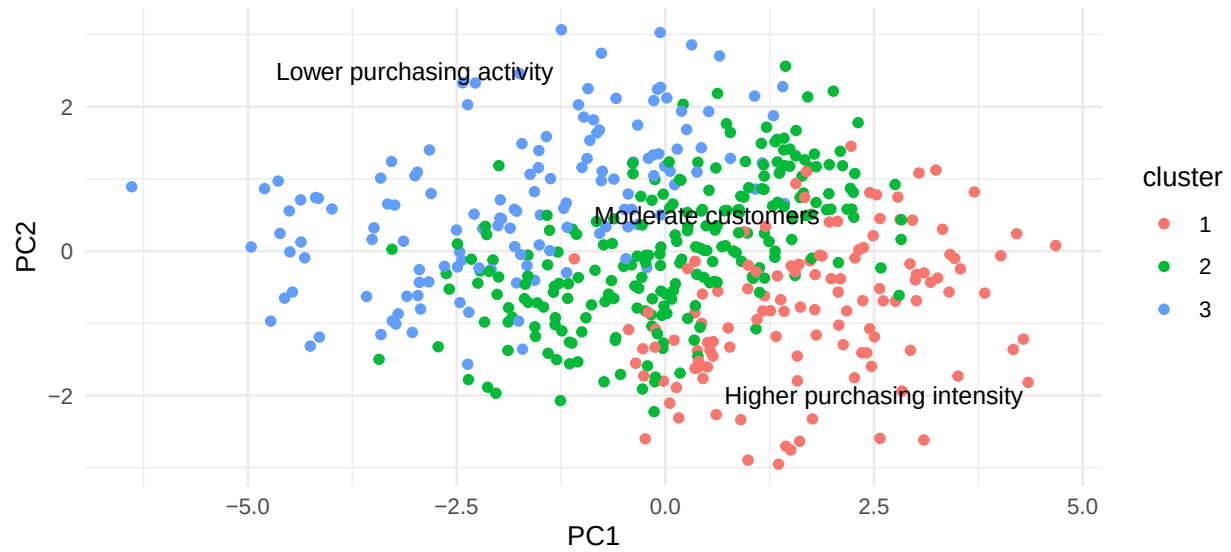


Figure 14: Scatter plot

4 Time series

We first plotted the three time series (ts1, ts2 and ts3) and visually assessed the stationarity of each. The plots for ts1 (15) and ts2 suggest they are stationary as they exhibit roughly constant mean and variances, without trend nor seasonality. However, the plot for ts3 suggests non-stationarity with evidence of trend. Formal unit-root test such as Augmented Dickey–Fuller (ADF) confirmed these: the p-values for ts3 is large (0.7322), so the null hypothesis of non-stationarity cannot be rejected, whereas the p-values for ts1 and ts2 are smaller than 0.01.

Although we are aware of the standard practice of inferring ARIMA parameters p and q from PACF and ACF, we find that they are not easy to apply when the time series have mixed AR and MA components. What we find easiest and most reliable is the following model selection procedure:

1. Use `auto.arima` to pick p , d , q parameters as a baseline model. The `auto.arima` function uses its own heuristics and use AIC (technically AICc, a version of AIC adjusted for small sample size, which in our case is not a concern) to measure model performance.
2. Manually fit ARIMA models for possible combinations of p and q (from 0 to 5 inclusive each), and consider the top 5 models ranked by AIC and BIC. This is to complement `auto.arima`, which does not actually consider all possible combinations of p and q .
3. Consider models with smaller BIC or AIC than the baseline `auto-arima` model. Since AIC as used by `auto-arima` tends to produce more complex models, we will try to find more parsimonious model by using BIC. (This is because BIC uses $k \cdot \ln(n)$ as a penalty term which for our sample size $n = 200$ is larger than $2k$ used by AIC). We have highlighted these candidate models in table 9 for ts1, table 10 for ts2 and table 9 for ts3.
4. Check the residuals for `auto-arima` and other candidate models from step 3. The residuals should look like a white noise because any pattern should be modelled away. This can be done by checking the ACF and PACF plots of the residuals, as well as by performing Ljung-Box test. The most parsimonious model with satisfactory residual is then selected.

Using the above steps, the selected final ARIMA model parameters are summarised in the table 8:

Table 8: Parameters of the final parameters

Series	Model	Intercept	ar1	ar2	ma1	ma2	ma3	Remarks
ts1	ARIMA(2,0,1)	0.642	0.891	-0.241	0.706	NA	NA	Better AIC and BIC than ARIMA(2,0,2)
ts2	ARIMA(2,0,3)	-0.017	-0.665	0.079	-0.499	0.19	0.61	From <code>auto.arima</code>
ts3	ARIMA(2,1,0)	NA	0.446	-0.535	NA	NA	NA	From <code>auto.arima</code>

The model residual diagnostics for the chosen models for each time series are all satisfactory:

- The residuals ACF and PACF display no significant spikes outside the confidence bounds, suggesting that no substantial autocorrelation remains in the residuals, as shown in figure 16 for ts1, figure 19 for ts2, and 22 for ts3.
- The residuals are approximately normally distributed, as can be shown from Q-Q plots in figure 17 for ts1, figure 20 for ts2, and figure 23 for ts3.
- The Ljung-Box tests indicate that we can reject the null hypothesis that the residuals are serially correlated for lag values up to 20, as can be shown in figure 18 for ts1, figure 21 for ts2, and figure 24 for ts3.

4.1 Figures and tables for Q4

Table 9: Top 5 ARIMA models for time series 1 ranked by AIC (left) and BIC (right). Highlighted in green is the final model selected over ARIMA(2,0,3) suggested by auto.arima, highlighted in yellow

p	d	q	AIC	BIC	p	d	q	AIC	BIC
2	0	3	550.860	573.948	2	0	1	554.022	570.513
2	0	4	552.701	579.087	1	0	2	554.119	570.611
3	0	3	552.819	579.206	1	0	1	559.434	572.627
4	0	2	552.890	579.277	2	0	3	550.860	573.948
5	0	5	553.855	593.435	2	0	2	555.449	575.239

Table 10: Top 5 ARIMA models for time series 2 ranked by AIC (left) and BIC (right). The models highlighted in red are checked for their residuals, since they have lower BIC than ARIMA(2,0,3) suggested by auto.arima (AIC=640.74, BIC=663.83), which is not in the table. But in all these candidate models, the residuals exhibit autocorrelation according to Ljung-Box test, so none is selected as the final model.

p	d	q	AIC	BIC	p	d	q	AIC	BIC
2	0	4	637.204	663.591	0	0	2	638.011	651.204
4	0	5	637.619	673.901	1	0	2	639.797	656.289
3	0	5	637.663	670.646	0	0	3	639.816	656.308
0	0	2	638.011	651.204	2	0	2	641.316	661.106
3	0	4	638.479	668.163	0	0	4	641.493	661.283

Table 11: Top 5 ARIMA models for time series 3 ranked by AIC and BIC, The final selected model ARIMA(2,1,3) is highlighted in green, which is also the model selected by auto.arima

p	d	q	AIC	BIC	p	d	q	AIC	BIC
2	1	0	555.135	565.015	2	1	0	555.135	565.015
2	1	5	556.704	583.051	2	1	1	556.830	570.003
2	1	1	556.830	570.003	3	1	0	556.851	570.024
3	1	0	556.851	570.024	3	1	1	557.974	574.441
5	1	2	557.587	583.934	0	1	3	561.945	575.119

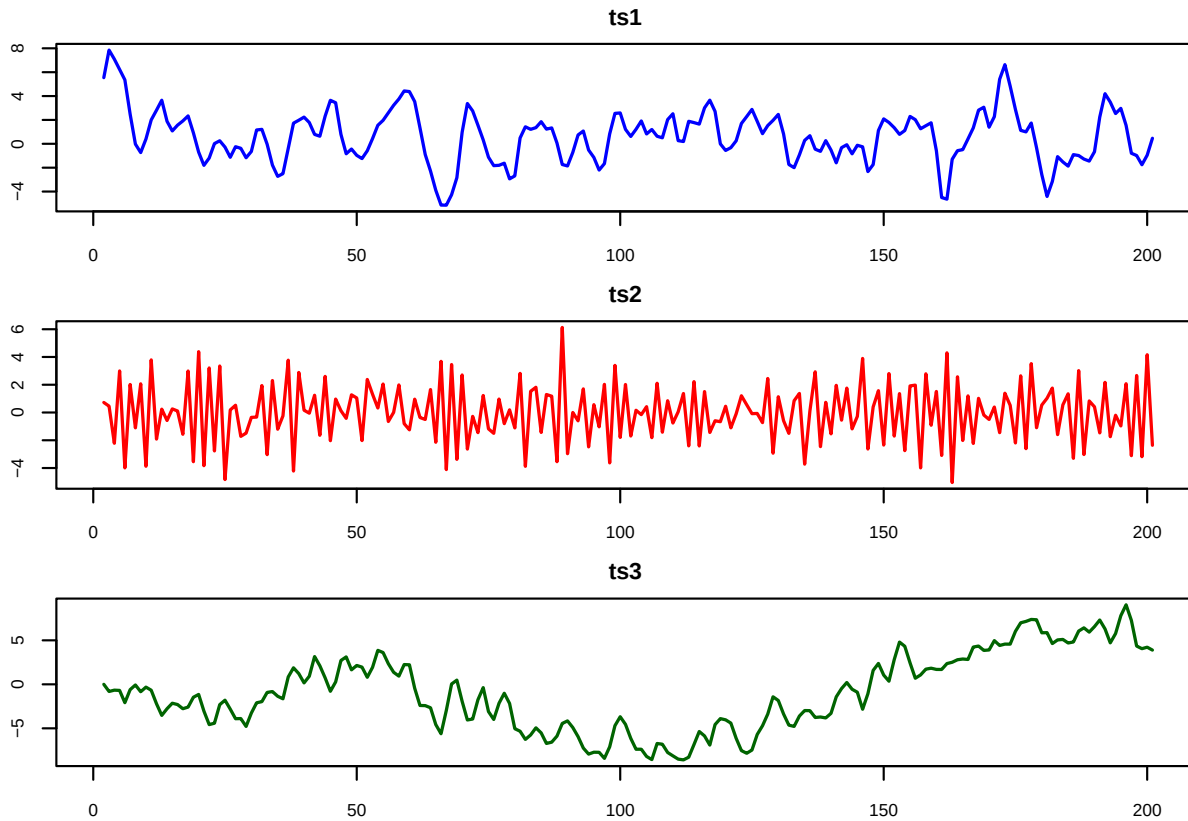


Figure 15: Time plots of the three simulated series

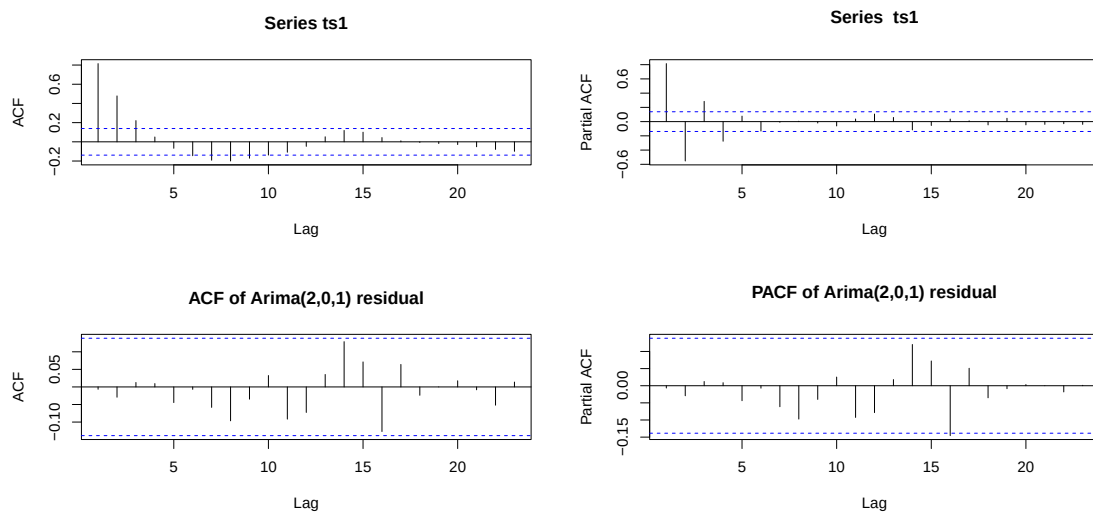


Figure 16: ACF vs PACF of ts1 (left), and of the residuals of fitted ARIMA(2,0,1) (right)

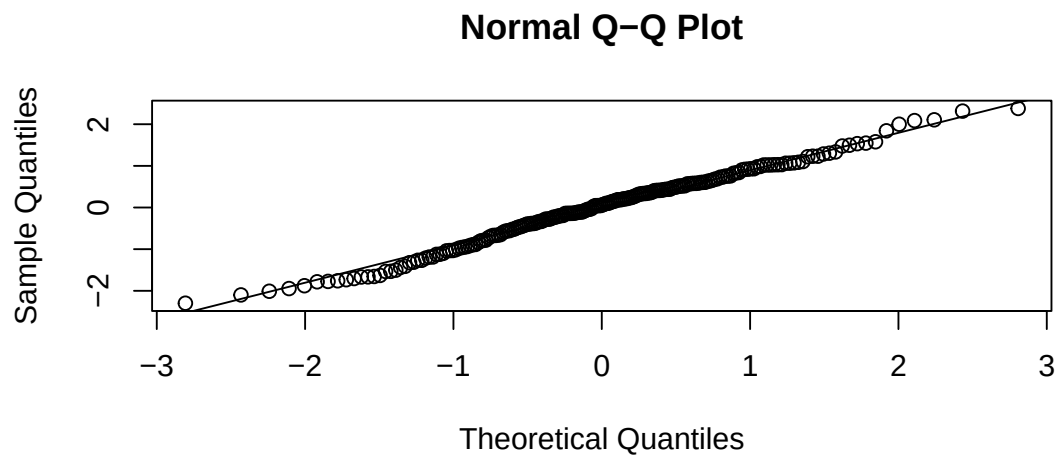


Figure 17: Q-Q plot of the residuals of fitted ARIMA(2,0,1) model for ts1

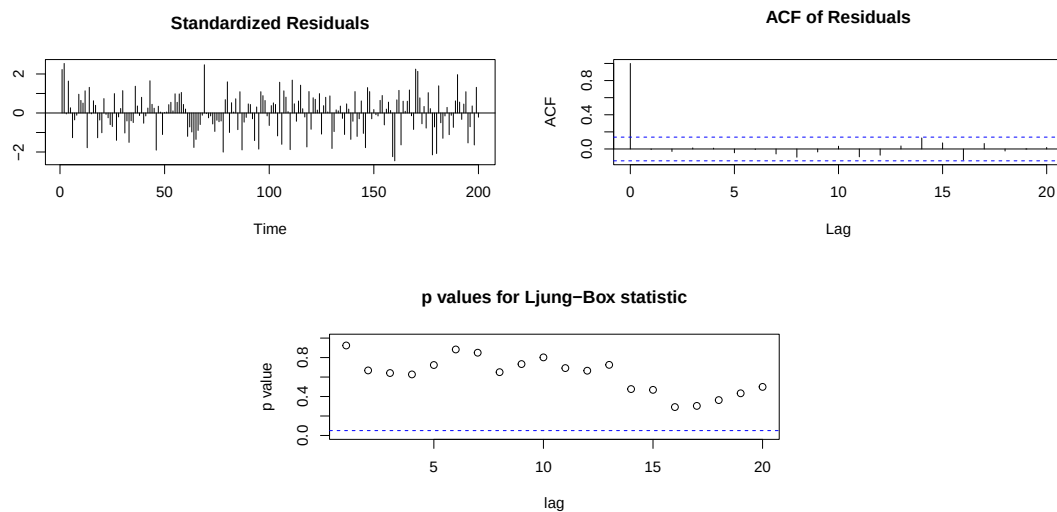


Figure 18: Diagnostic plots for the fitted ARIMA(2,0,2) model for ts1, including standardized residuals, residual ACF, and Ljung-Box test results.

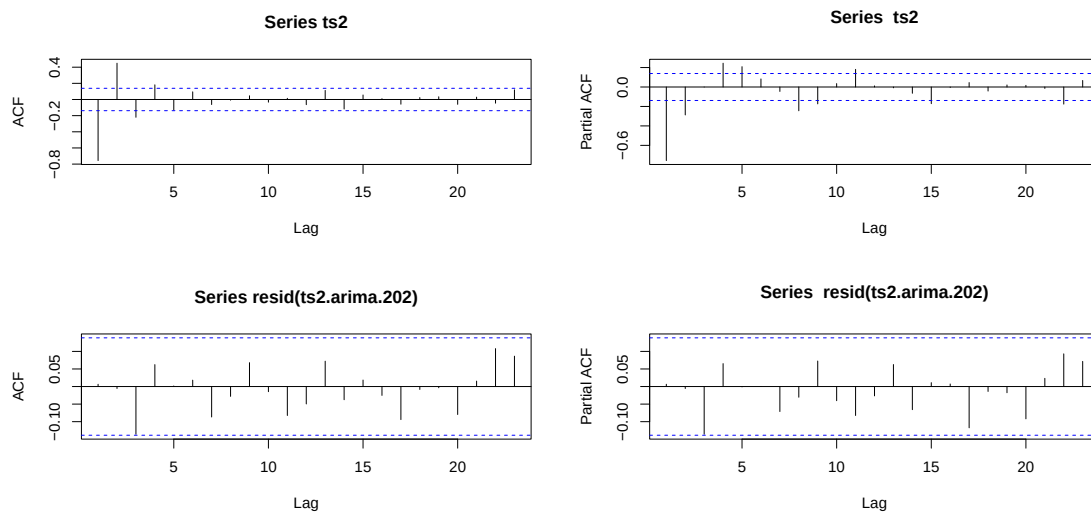


Figure 19: ACF vs PACF of *ts2* (top), and of the residuals of fitted ARIMA(2,0,3) (bottom)

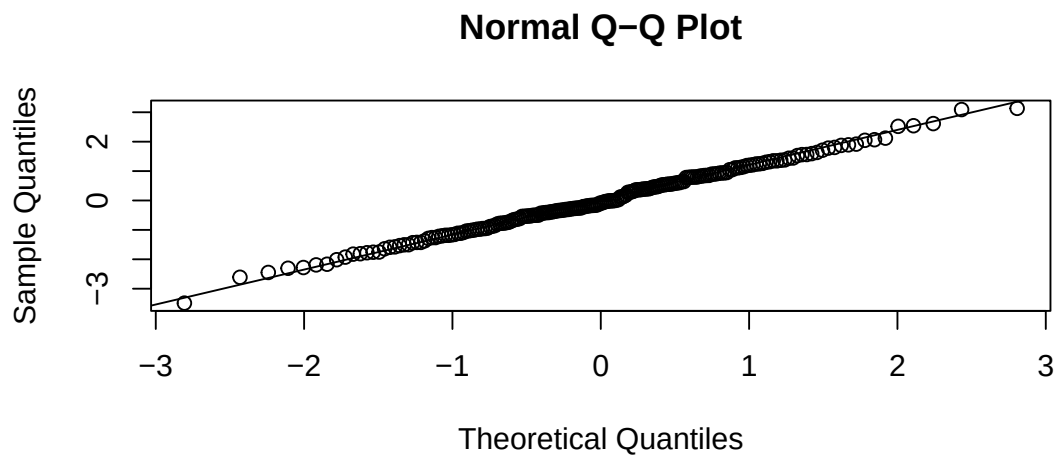


Figure 20: Q-Q plot of the residuals of fitted ARIMA(2,0,2) model for *ts2*

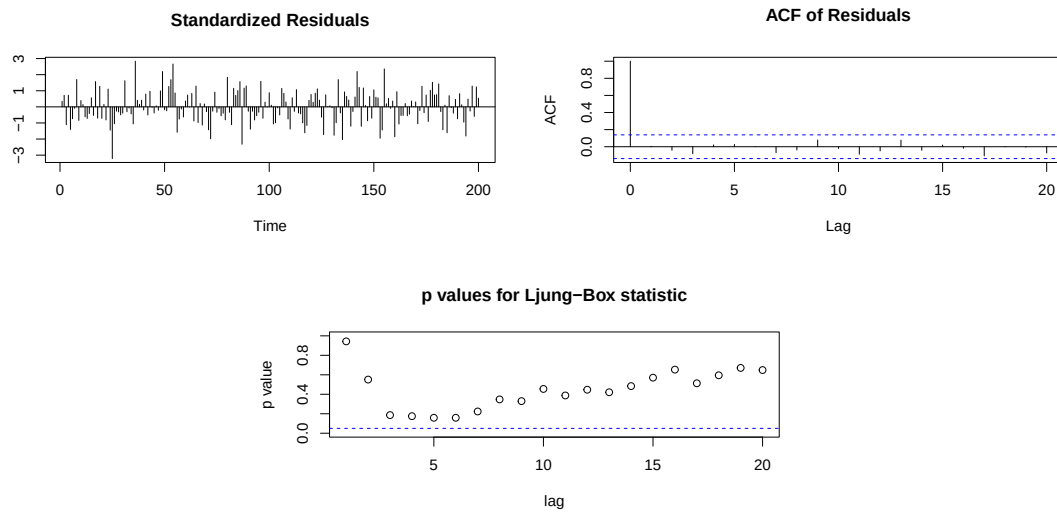


Figure 21: Diagnostic plots for the fitted ARIMA(2,0,2) model for ts_2 , including standardized residuals, residual ACF, and Ljung-Box test results.

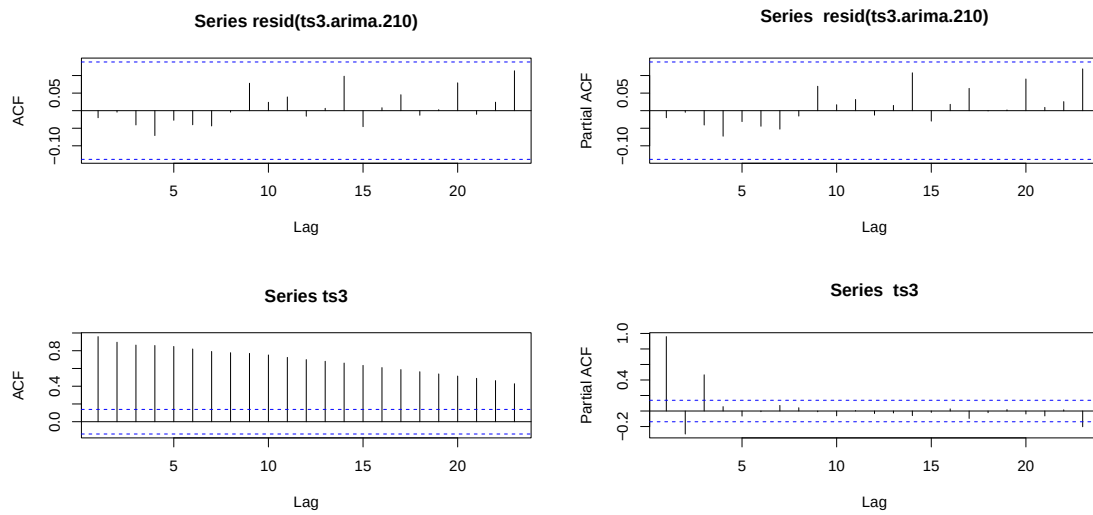


Figure 22: ACF vs PACF of ts_3 (top), and of the residuals of fitted ARIMA(2,1,0) (bottom)

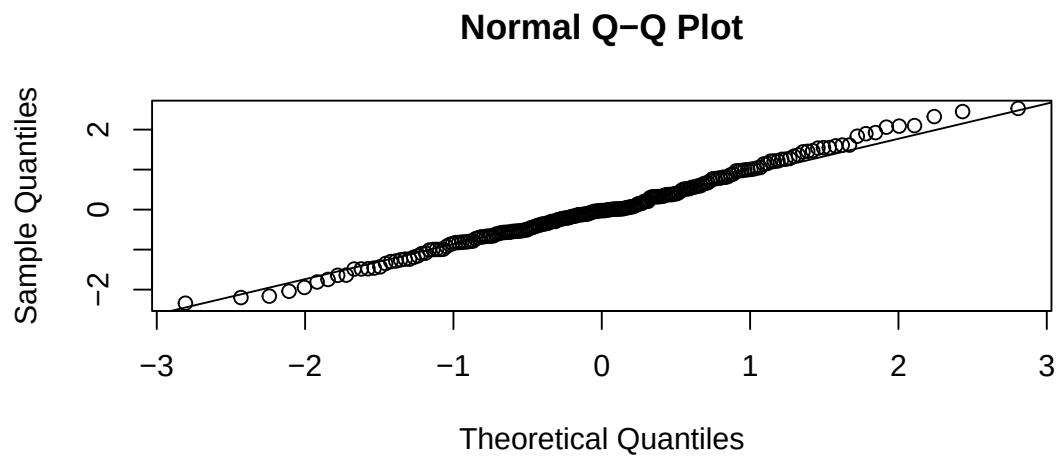


Figure 23: Q-Q plot of the residuals of fitted ARIMA(2,1,0) model for ts3

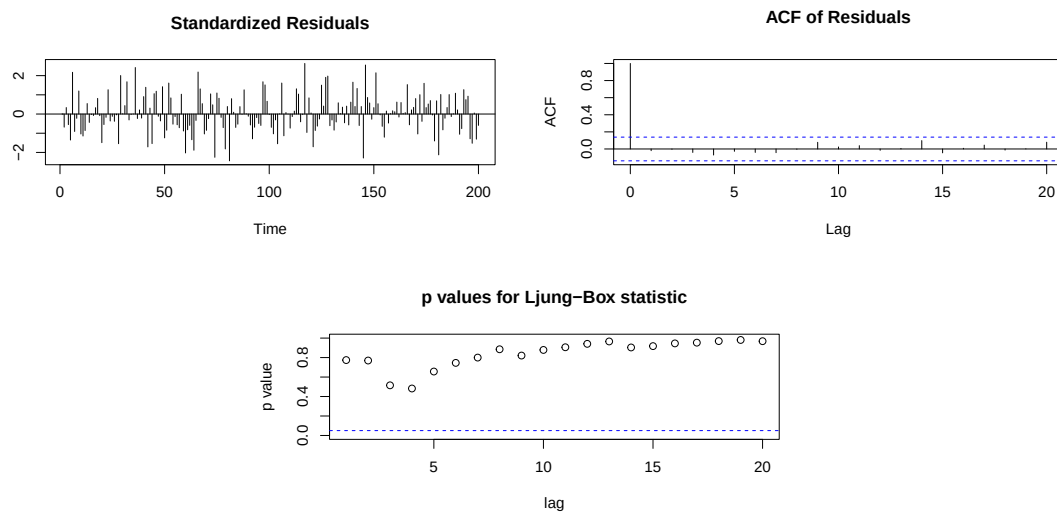


Figure 24: Diagnostic plots for the fitted ARIMA(2,1,0) model for ts3, including standardized residuals, residual ACF, and Ljung-Box test results.

5 Appendix

5.1 bonus : Q1 Factor analysis comparison(number of factors 5 vs 6)

The heatmaps compare the rotated factor loading structures obtained from the five-factor and six-factor solutions using varimax rotation. In the five-factor solution, items generally load strongly on a single factor, producing a relatively clear and interpretable structure. When a sixth factor is introduced, several items show weaker or more dispersed loadings across factors, suggesting a less stable structure. In particular, the additional sixth factor captures only a limited number of items and does not form a clearly interpretable dimension. Therefore, the five-factor solution appears more parsimonious and provides a more coherent representation of the underlying personality traits.

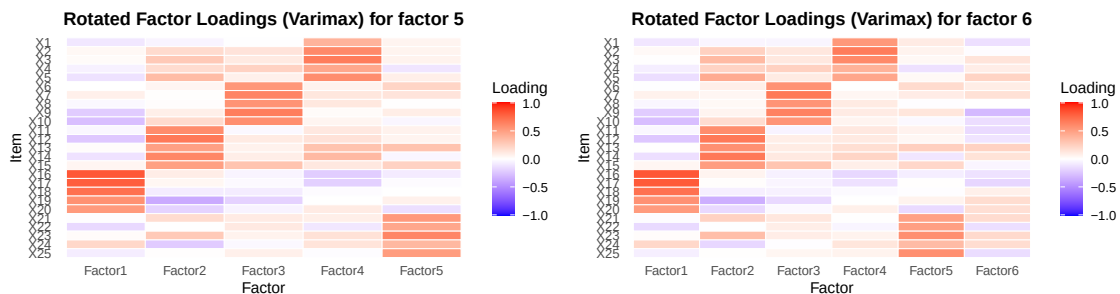


Figure 25: Comparison of the number of factors (5 vs 6)

The rotated factor loading tables allow a comparison between the five-factor and six-factor solutions. In the five-factor model, the items cluster clearly into five interpretable groups, with most salient loadings above 0.40 concentrated on a single factor. In contrast, the six-factor solution does not substantially improve the structure, as the additional sixth factor (F6) does not show any strong loadings above the threshold. Most items continue to load on the same factors as in the five-factor solution, indicating that the extra factor does not capture a distinct dimension. Therefore, the five-factor model appears more parsimonious and provides a clearer and more interpretable representation of the underlying personality structure.

Table 12: Rotated Factor Loadings for 5 Factors ($|\text{loading}| \geq 0.40$)

Factor	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	X12	X13	X14	X15	X16	X17	X18	X19	X20	X21	X22	X23	X24	X25
F1																0.82	0.79	0.71	0.56	0.52					
F2											0.59	0.67	0.49	0.61	0.49										
F3						0.53	0.62	0.55	0.65	0.57															
F4		0.60	0.66	0.45	0.58																				
F5																					0.52	0.45	0.61		0.51

Table 13: Rotated Factor Loadings for 6 Factors ($|\text{loading}| \geq 0.40$)

Factor	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	X12	X13	X14	X15	X16	X17	X18	X19	X20	X21	X22	X23	X24	X25
F1																0.82	0.81	0.71	0.56	0.51					
F2					0.44						0.58	0.67	0.57	0.67	0.51										
F3						0.56	0.68	0.55	0.64	0.55															
F4		0.53	0.66	0.60	0.40	0.46																			
F5																					0.48	0.49	0.57		0.57
F6																									

5.2 bonus : Q2 Copula

5.2.1 Modelling using Poisson and Tweedie margin

The first approach retains all observations and models claim amounts using a Tweedie distribution, which accommodates mixed distributions with probability mass at zero. However, this approach could not be implemented because the current GJRM library does not support joint regression combining Poisson frequency with Tweedie severity margins. The second approach truncates zero-claim observations and models strictly positive claim amounts using alternative continuous distributions. This is also consistent with actuarial practice, where the focus is typically on modelling positive claims for pricing and reserving purposes. The remainder of Question 2 therefore adopts the second approach.

5.2.2 Goodness-of-fit tests

Goodness-of-fit tests were conducted using the Kolmogorov–Smirnov (KS) and Cramér–von Mises (CvM) statistics. Null hypothesis is data follow the fitted marginal distribution. The KS statistic ($D_1=0.93707$, $D_2=0.74048$) measures the maximum difference between the empirical and theoretical cumulative distributions of zero truncated poisson and inverse gaussian, while the Cramér–von Mises statistic ($\omega_1^2 = 1268.4$, $\omega_1^2 = 653.09$) measures the overall squared discrepancy across two distributions. Both tests reject the null hypothesis that the observed data follow the fitted distributions exactly ($p\text{-value} < 2.2 \times 10^{-16}$). However, with a sample size of $n = 4624$, these tests are highly sensitive and may detect even minor deviations from the theoretical distribution.

5.2.3 Adding covariates into the models

Additional models including covariates were explored to assess whether observable policy characteristics improve the marginal model specification. In particular, vehicle value, geographic area and gender were considered, as these variables may plausibly influence accident risk and claim costs. For example, vehicle value may affect repair costs, while area and gender may capture differences in driving behaviour or traffic exposure. In the claim frequency model, the intercept-only truncated Poisson specification achieves the lowest AIC and BIC compared to models with extra covariates. For the claim severity model, the inclusion of 4 extra covariates, including vehicle value, area and gender producing the lowest AIC. However, BIC again favours the simpler intercept-only specification. Overall, it is suggested that the additional covariates provide limited explanatory power in this dataset.

5.3 bonus : Q3 Principal Component Analysis

The PCA biplot illustrates both the observations and the directions of the original variables in the space of the first two principal components. Variables related to purchasing behaviour, such as income, brand loyalty, shopping frequency, and spending score, are largely aligned with the direction of PC1, indicating their strong contribution to this component. In contrast, age is oriented in a different direction, suggesting that it captures a different aspect of variation in the data. Overall, the biplot confirms that purchasing-related variables primarily drive the variation explained by the first principal component.

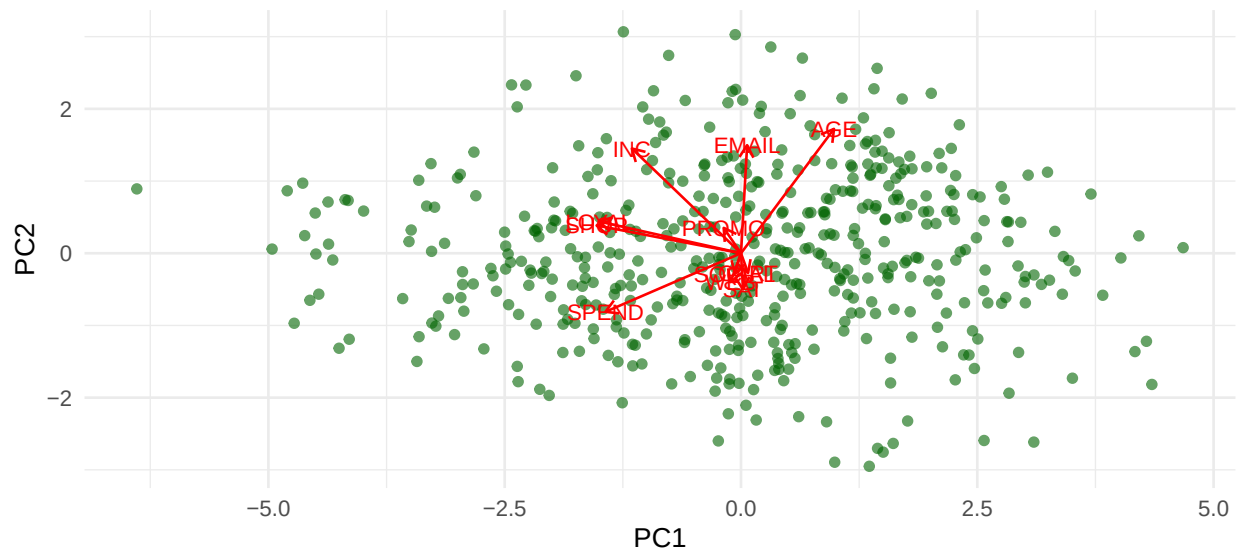


Figure 26: PCA Biplot

The three-dimensional PCA score plot provides a visual representation of the observations across the first three principal components. The points form a continuous distribution rather than clearly separated clusters, indicating that customer behaviour varies gradually across the dataset. This visualization further supports the interpretation that the main variation in the data is captured by the first few principal components.

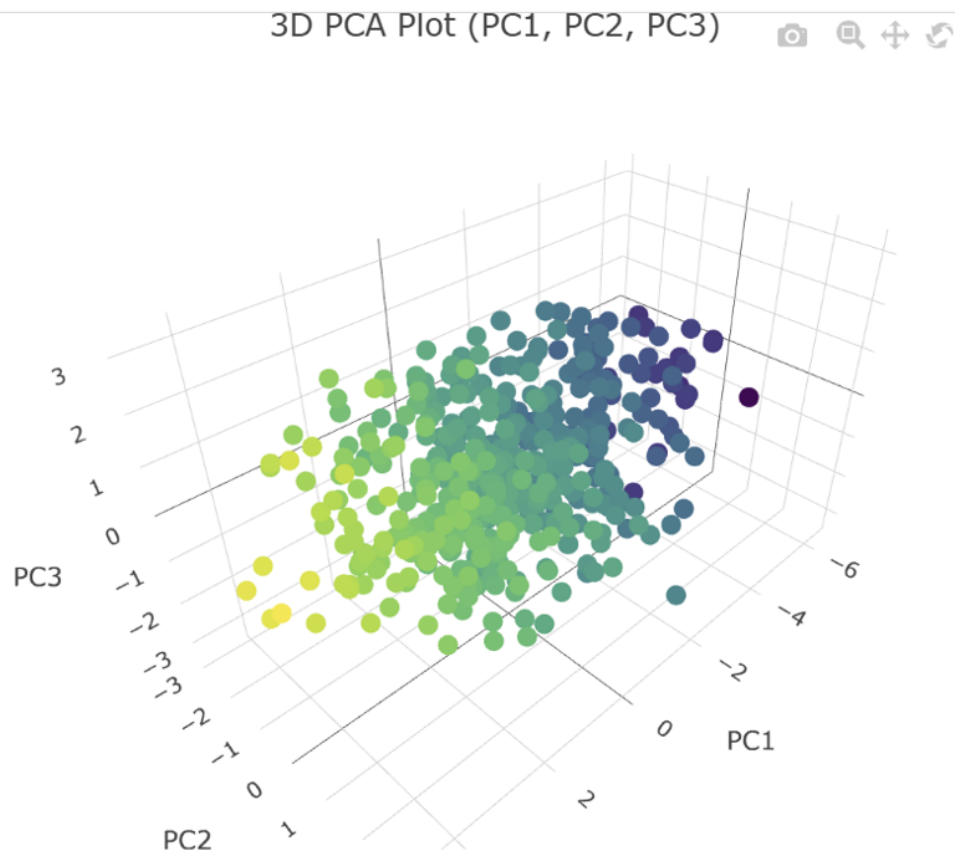


Figure 27: 3D PCA score plot of the first three principal components.